Ontology-Driven Triple-to-Text Generation Using LLaMA for University Knowledge Graphs

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***Abstract*—Converting structured Resource Description Framework (RDF) triples into fluent natural language plays a crucial role in improving the accessibility and interpretability of knowledge graphs. This paper presents a domain-adaptive, ontology-driven pipeline for verbalizing triples extracted from a university knowledge graph into coherent sentences. The proposed multi-stage framework includes semantic filtering, fragment parsing, grammatical refinement, and generation using a locally hosted LLaMA language model with domain-specific prompts. Experimental evaluation, conducted on triples from an academic institutional ontology, shows strong semantic alignment with reference texts. The results demonstrate that the system effectively preserves the intended meaning of triples while introducing lexical variation, though occasional predicate shifts, entity omissions, and hallucinated content remain. These insights pave the way for targeted refinements in factual consistency and lexical precision, thereby enhancing the reliability of domain-specific triple-to-text generation for downstream applications such as semantic search and question answering.**

***Index Terms*—RDF verbalization, knowledge graph, LLaMA, natural language generation, ontology-driven processing.**

I. Introduction

In the era of knowledge-intensive applications, the ability to convert structured data into human-readable natural language plays a pivotal role in enhancing machine interpretability and user accessibility. This process—commonly referred to as data-to-text generation—has seen significant advances across domains which uses semantic web technologies [1], [2]. Among the various forms of structured input, RDF triples (⟨subject, predicate, object⟩) derived from Knowledge Graphs (KGs) represent a foundational data structure for encoding semantic relationships between entities. The task of triple-to-text generation, a subfield of natural language generation (NLG) [3], focuses specifically on transforming these RDF triples into coherent, grammatically correct, and contextually appropriate text. This conversion not only facilitates human understanding of linked data but also enables downstream applications such as semantic search, chatbots, ontology verbalization [4], KG summarization [5]. Earlier research in this area explored rule-based and template-driven approaches [6], which, while interpretable, lacked scalability and adaptability. With the emergence of neural sequence-to-sequence models discussed in Section II , a shift toward data-driven methods allowed for greater linguistic variation and contextual awareness.

Despite significant progress in natural language generation, generating fluent and semantically accurate natural language descriptions from structured RDF triples remains a challenging task, especially when working with domain-specific knowledge graphs such as those in academic institutions. Most existing data-to-text systems either rely heavily on hand-crafted templates or require large-scale annotated datasets for training [7]. Moreover, they often struggle to maintain factual consistency when converting custom URIs and domain-specific relations into coherent sentences [8]. This leads to a gap in effective sentence generation from sparse or customized triples derived from educational knowledge graphs, which hampers downstream applications like domain-specific question answering.

The primary objective of our research work is to develop an domain-adaptive framework for generating coherent and factually accurate natural language sentences from RDF triples extracted from a university knowledge graph.

The structure of the paper is organized into several key sections to comprehensively address the research objectives. It begins with an exploration of studies relevant to the research topic. Following this, the methodology section outlines the process of extracting subject–predicate–object triples from a custom university knowledge graph. Subsequently, the paper delves into the development, and training of the proposed model followed by a comprehensive analysis of the obtained results. Finally, the paper concludes with a future work of the findings.

II. Related Work

Despite the advances from hand-written rules to sequence to sequence models, challenges remain in producing coherent, contextually appropriate text with minimal supervision. In the following literature survey, we review and compare key methods for triple-to-text generation, outlining their innovations and open research problems.

The work [10] proposes a novel Triple-to-Text (T2T) framework that improves the quality of natural language generation from RDF triples by optimizing the inverse KL divergence, effectively reducing low-quality sentence generation. While the method shows promising improvements over existing baselines on multiple datasets, it may still face some challenges due to approximations of the real distribution, which could affect performance on highly complex or diverse data. The authors in research work [6] use a rule-based approach to convert RDF triples into natural language text. Being rule-based approach, it may suffer from scalability issues and limited adaptability to diverse or large-scale datasets. In [11] combining Byte Pair Encoding in encoder-decoder transformer , learned embeddings, and back-translation leads to state-of-the-art or competitive performance on RDF-to-Text generation. While back-translation improves performance, it increases training complexity and computational cost.

In [12], two layered bidirectional encoder-decoder model with LSTM units along with attention mechanism is used to generate sentences from RDF triples. While the proposed enhancements improve text generation quality, there is still room for further refinement in handling complex semantic structures. The key idea in [13] is to use a triple encoder that better captures the semantic structure of triples, and feed this into an LSTM-based decoder to generate text. While effective, the model may face scalability challenges when dealing with dense or intricate RDF structures. FRED-T5 is explored in [14] to convert triples of KG to sentence.

Authors in [15] have manually designed templates for converting RDF triples into Arabic sentences. Arabic morphological analyzers and generators are used to ensure grammatical correctness. While the proposed approach demonstrates promising results, a more extensive comparison with state-of-the-art neural models would further strengthen the evaluation and contextualize its performance.

The research work [16] combines Gated Graph Neural Networks (GGNN) with a Bidirectional GRU to encode both graph structure and sequential information of RDF triples. The work introduce the fusion mechanism to combine outputs from the GGNN and the BiGRU, leading to more context-aware encodings for text generation.

The research work in [18] proposes a dual learning framework in which triples are converted into sentences and back using graph transformer The research work in [17] proposes a new evaluation metric called Entity-Based Semantic Adequacy to assess data-to-text generation systems.

III. Methodology

We propose a five-stage pipeline for converting structured RDF triples into fluent natural-language sentences as shown in Fig. 1. Formally, let

$$\mathcal{G} = (V, E)$$

be our knowledge graph, where V is the set of entities (nodes) and E the set of labeled relations (edges). The stages are:

A. Triple Extraction

We define a retrieval function

$$R : \mathcal{G} \;\rightarrow\; \mathcal{T},$$

where

$$\mathcal{G} = (V, E) \quad \text{and} \quad \mathcal{T} = \{(s, p, o) \mid s, o \in V,\; p \in E\}.$$

Here, $R(\mathcal{G})$ simply enumerates all RDF triples in the graph for downstream processing.

B. Semantic Filtering

In this stage, we discard not only triples of low semantic relevance but also those included solely to satisfy the knowledge graph's syntactic schema (e.g., `rdf:type` and `rdfs:label` declarations). Formally, let

$$\begin{aligned}\mathcal{N} = \{&\texttt{http://www.w3.org/1999/02/22-rdf-syntax-ns\#},\\ &\texttt{http://www.w3.org/2000/01/rdf-schema\#},\\ &\texttt{http://www.w3.org/2002/07/owl\#},\\ &\texttt{http://www.w3.org/2001/XMLSchema\#}\}.\end{aligned}$$

We then filter by predicate namespace, hence

$$\mathcal{T}' \;=\; \big\{(s, p, o) \in \mathcal{T} \;\big|\; p \notin \mathcal{N}\big\}.$$

C. Fragment Parsing

Each retained triple $t = (s, p, o) \in \mathcal{T}'$ is mapped to a raw text-fragment triple via

$$\phi : \mathcal{T}' \;\rightarrow\; \mathcal{F}, \quad \phi(s, p, o) = (f_s,\, f_p,\, f_o),$$

where f_s, f_p, f_o are the IRI's final segment of subject, predicate, and object.

D. Fragment Refinement

We apply a refinement function

$$\eta : \mathcal{F} \;\rightarrow\; \mathcal{F}^*$$

to enforce grammatical agreement and normalize syntax. Thus

$$\mathcal{F}^* = \{\eta(f_s, f_p, f_o) \mid (f_s, f_p, f_o) \in \mathcal{F}\}.$$

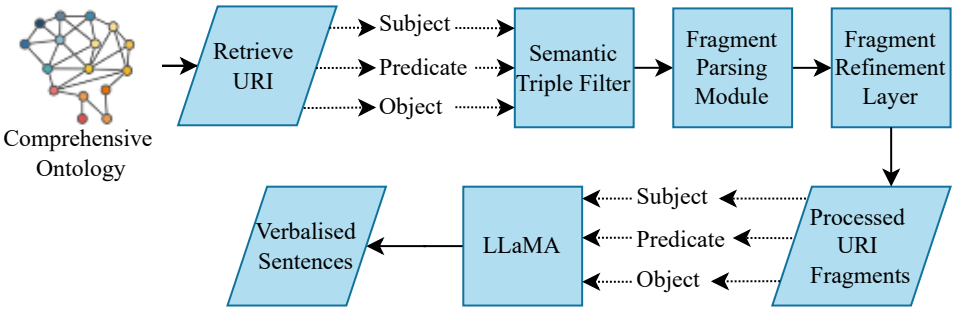

Fig. 1. A comprehensive framework for the seamless verbalization of structured RDF triples into articulate natural-language sentences.

E. Conditional Language Model: LlaMa

LLaMA defines a conditional language model by parameterizing a probability distribution over output token sequences given an input sequence. Formally, the probability of generating a sequence of output tokens $y_{1:M} = (y_1, y_2, \ldots, y_M)$ conditioned on an input sequence $x_{1:N} = (x_1, x_2, \ldots, x_N)$ is defined as:

$$p_\theta(y_{1:M} \mid x_{1:N}) = \prod_{t=1}^{M} p_\theta(y_t \mid y_{<t}, x_{1:N}), \tag{1}$$

where θ denotes the model parameters. At inference time, the model generates the output sequence $\hat{y}$ that maximizes this conditional probability:

$$\hat{y} = \arg\max_{y \in V^*} p_\theta(y \mid x), \tag{2}$$

where V^* is the set of all possible token sequences over the vocabulary V.

Following the formulation above, our system leverages this conditional distribution to generate coherent natural language sentences from structured RDF-style triples. The input to the model is a serialized representation of a triple $x = (s, p, o)$, comprising a subject, predicate, and object. The output $\hat{y}$ is a well-formed sentence that semantically expresses the relationship encoded in the input.

To operationalize this, we employ a locally hosted version of the LLaMA model via LMStudio. The model is accessed through an API interface using domain-specific system prompts π_d, which are fixed instructions that guide the model to produce faithful and fluent realizations of the input triples. Each API request serializes the input triple and combines it with the corresponding system prompt, forming the input context $x_{1:N}$ to the model.

Formally, the prompt-augmented input is defined as:

$$x_{1:N} = [\pi_d \parallel \texttt{Subject}, \texttt{Predicate}, \texttt{Object}],$$

where $\parallel$ denotes string concatenation and π_d is the domain-specific instruction.

During generation, the model samples or decodes the most probable output sentence $\hat{y} \in V^*$ that maximizes the conditional probability in Equation (2). In practice, we use greedy or temperature-controlled decoding strategies provided by the LMStudio interface. The decoding parameters used for sentence generation are summarized in Table I.

TABLE I
DECODING PARAMETERS FOR ENHANCED GENERATION

Parameter	Configured Setting
`max_tokens`	150
`top_p`	0.9
`repetition_penalty`	1.2
`stop`	`["\n", "END"]`

The generated output is then parsed and stored along with the original triple, resulting in an enriched dataset of the form:

`Subject,Predicate,Object,Generated Sentence.`

This approach enables scalable, automated generation of fluent, human-readable text from structured semantic representations, and serves as the foundation for building interpretable knowledge graph applications and semantic QA systems.

F. Results and Analysis

In this work, we propose a multi-stage, ontology-driven pipeline as shown in Figure 1 that converts raw RDF triples into fluent, end-user–ready sentences. Table II presents representative subject–predicate–object triples and the corresponding sentences generated by our system.

Table III reports the automatic and human evaluation results for our generated sentences. Our system shows strong semantic alignment with the references: Sentence-BERT cosine similarity is 0.85 $\pm$ 0.09 indicating that generated sentences occupy a very similar embedding space to the gold text. In contrast, surface-form overlap is only moderate: BLEU-4 = 0.25, ROUGE-1 = 0.55, and METEOR = 0.45. This pattern suggests the model preserves meaning but often paraphrases rather than reproducing exact n-grams.
Human ratings corroborate this finding: **Fluency** $= 3.0/5$ (grammatical but sometimes stiff), **Relevance** $\approx 4.0/5$ (faithful to the input triple), and **Coherence** $= 3.5/5$ (mostly consistent with occasional awkward links). Taken together, automatic metrics that reward semantics (e.g., SBERT, METEOR) align better with human judgments than n-gram–heavy BLEU-4, highlighting that the system captures content well while leaving room to polish phrasing.

TABLE II
SAMPLE TRIPLES AND THEIR GENERATED SENTENCES

Subject	Predicate	Object	Generated Sentence
CCE	Total Eligible Students	117	The CCE was a total offer of 112.
CCE	Total Placed	107	In the CCE, a total of 107 students were placed.
Joint Cultural Secretary	type	Student Council	The joint cultural secretary has to work under the student council.
block_12	type	girls	Block 12 is a type of hostel for girls.
block8	elevator	FALSE	The elevator is located in block8.
block_14	gym	TRUE	There is gym in block 14.

TABLE III
EVALUATION METRICS AND THEIR RESULTS FOR GENERATED SENTENCES.

Metric	Description	Value
Automatic metrics		
Cosine similarity using Sentence-BERT(all-MiniLM-L6-v2)	Computes the semantic closeness of two sentences by measuring the cosine of the angle between their fixed-length embeddings	0.85 ± 0.9
BLEU-4	n-gram overlap score against reference sentences	0.25
ROUGE-1	Longest common subsequence measure (recall-oriented)	0.55
METEOR	Alignment-based F-score with synonym and stemming support	0.45
Human evaluation (Likert scale 1–5)		
Fluency	Grammaticality and naturalness of sentences	3.0
Relevance	Correct representation of the input triple's meaning	4.0
Coherence	Logical consistency and contextual fit	3.5

IV. ERROR ANALYSIS

To better understand the limitations of the proposed SPO-to-sentence generation system, we conducted a detailed error analysis of the outputs produced by the fine-tuned LLaMA model. The goal of this analysis is to identify the types of deviations from the intended meaning and structure, as defined by the subject–predicate–object (SPO) triples, and to inform targeted improvements in future iterations of the model. Errors were manually inspected and categorized into semantic (meaning shift), lexical (wording changes), structural (subject/object omission), hallucination (addition of unsupported content), and grammatical/style issues. Table IV and V illustrates an example case and the corresponding error categories.

V. CONCLUSION AND FUTURE SCOPE

This study presented a domain-adaptive, ontology-driven framework for converting RDF triples from a university knowledge graph into fluent natural language sentences. The proposed multi-stage pipeline, integrating semantic filtering, fragment parsing, refinement, and generation using a locally hosted LLaMA model, demonstrated strong semantic alignment with reference texts, as indicated by both automatic and human evaluations. While the system effectively preserved the intended meaning of triples and offered lexical variation, a particular challenge of controlled generation methods to keep triple-to-text output factually correct still exists.

For future work, fact-checking modules can help mitigate hallucinations. Also, context-aware generation that considers multiple related triples jointly, rather than in isolation, may lead to more coherent and informative sentence outputs. Finally, since the current experiments are based on a custom dataset, a comparative study with standard benchmark datasets remains an important future direction to better assess the system's generalizability and performance against existing approaches.

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TABLE IV

ERROR ANALYSIS FOR GENERATED SENTENCE FROM SPO TRIPLE. **SPO:** (STUDENT TEACHING STUDENT PROGRAMME, TYPE, MAJOR EVENTS PROGRAMMES). **GENERATED SENTENCE:** "IN OUR PROGRAMMES, THE STUDENT GETS AN OPPORTUNITY TO PARTICIPATE IN MAJOR TYPE OF EVENTS."

Error Type	**Observation**	**Explanation**
Meaning shift (semantic error)	Predicate `type` expressed as "participate in"	The triple expresses a classification (*is a*), but the generated sentence changes the relation to participation, altering the intended meaning.
Subject omission	"Student Teaching Student Programme" not mentioned	The generated sentence replaces the subject with the generic phrase "the student," losing the original entity reference from the triple.
Object alteration	"Major events Programmes" $\rightarrow$ "major type of events"	The object label is paraphrased and partially distorted, losing its specific taxonomic meaning.
Hallucination	Added "our programmes" and "opportunity"	These phrases are not supported by the SPO triple and introduce unrelated content.
Fluency/coherence issue	Sentence structure is verbose and indirect	The generated sentence uses a long, roundabout construction

TABLE V

ERROR ANALYSIS FOR THE GENERATED SENTENCE FROM THE TRIPLE (*block_12, type, girls*).

Error Category	**Description**
Semantic Hallucination	The generated sentence adds extra information – "offers a comfortable living space" – which is not present in the given triple. This introduces content that cannot be validated from the RDF data.
Predicate Mis-verbalization	The predicate *type* is verbalized as "caters to", which alters its intended meaning. The correct mapping should be "is a" or "is designated for" to maintain semantic fidelity.
Redundancy / Wordiness	The phrase "for the type of students it caters to, which are girls" is unnecessarily long and convoluted, reducing clarity.
Entity Surface Form	The subject is presented as *block_12*, which is machine-readable. It should be converted to its natural language label "Block 12" for better readability.

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